

Limitations of Anonymous, Self-Submitted Data Sources

Module 3 Written Reflection — Grad School Café Analysis

One of the most fundamental limitations of using a platform like Grad Café as a data source is that it relies entirely on voluntary, self-reported submissions from anonymous users. Because participation is optional and there is no verification mechanism, the resulting dataset is subject to significant selection bias. Applicants who choose to submit their information may systematically differ from those who do not: individuals with particularly strong credentials, those admitted to prestigious programs, or those engaged enough in the applicant community to share their outcomes are more likely to post. This explains, at least partially, why the average GRE Quantitative score in our dataset (≈ 165) substantially exceeds the official ETS average of 157 for the 2022–2023 testing year. The platform is not capturing a random cross-section of all applicants; it is capturing a self-selected group that skews toward high performers and selective institutions. Consequently, any averages or percentages we compute — acceptance rates, mean GPAs, GRE distributions — cannot be interpreted as representative of the true applicant population. Treating them as such would lead to systematically overestimating applicant strength and admission difficulty, misleading prospective students who rely on this information to calibrate their own chances.

A second category of limitations stems from data quality issues inherent to anonymous, free-text submission. Users may enter information inconsistently (e.g., 'CMU,' 'Carnegie Mellon,' and 'Carnegie Mellon University' for the same institution), omit fields entirely, or even submit fabricated or exaggerated entries, since there is no accountability mechanism. Our own analysis illustrated this directly: query results for Questions 8 and 9 differed depending on whether we used the scraped program fields or the LLM-normalized university fields, because the raw scraped text contained enough naming variation to suppress matches. Missing data is another serious concern — GPA, GRE, and GRE V values were absent for large portions of the dataset, meaning averages in Question 3 reflect only the subset of users who chose to report those metrics, likely introducing further upward bias. Beyond measurement error, the data contains no ground-truth labels: there is no way to confirm that a reported 'Accepted' status is genuine, or that the program listed actually corresponds to the institution named. These compounding issues — selection effects, inconsistent formatting, missingness, and lack of

verifiability — mean that while Grad Café provides a rich and interesting dataset for exploratory analysis, its findings should always be contextualized against official, audited statistics before drawing any policy or decision-relevant conclusions.